

Assignment 1: Polynomial Regression

Power Plant Optimization & Subterranean Reservoir Mapping

Student Roll Number: BT2024032 | Course: Machine Learning | Code Repository: <https://github.com/gagansyam/ML-Assignment-1>

1. Introduction

In this assignment, I solved two prediction problems as the Lead Renewable Energy Engineer for a geothermal power plant expansion project:

- Phase 1: Power Plant Steam Turbine Optimization (var1):** Predicting the plant's Net Power Score (y) using 6 operational control variables (x1 to x6) representing percentage deviations of high-pressure valves, coolant flow, re-injection pressure, blade pitch, gas exhaust, and steam inlet pressure.
- Phase 2: Subterranean Thermal Reservoir Mapping (var2):** Predicting the subsurface Thermal Anomaly Score (y) across 3D spatial coordinate offsets (x1, x2, x3) from our basecamp, helping the team select optimal high-yield drill sites.

My goal was to develop polynomial regression models that fit the true response surfaces without underfitting or overfitting. I evaluated all models using 5-fold cross-validation on the training data, tracking both Mean Squared Error (MSE) and R2 score.

2. Phase 1: Steam Turbine Optimization (var1)

2.1 Data Inspection

I checked both the training set (BT2024032_train_var1.csv) and test set (BT2024032_test_var1.csv). Each contains 1,000 samples with no missing or null values. All 6 input features are normalized percentage deviations bounded in [-1.0, 1.0]. The target variable (Net Power Score) has a mean of 0.74, a standard deviation of 3.13, and ranges from -8.83 to +11.50. Linear correlations between individual inputs and the output were weak to moderate (highest was -0.26 for x5 and -0.23 for x3). This indicated that the turbine parameters do not act in isolation, and higher-order interaction terms are required to capture the power output.

2.2 Baseline Model (Degree 1)

I began with a simple linear regression model (Degree 1). Using 5-fold cross-validation, it yielded:

- CV Validation MSE:** 8.5302
- CV Validation R2:** 0.1248 (only 12.5% of variance explained)

This confirmed severe underfitting (high bias). A linear plane cannot model the aerodynamic and thermodynamic interactions inside a multi-stage steam turbine.

2.3 Polynomial Degree Exploration & Regularization

I expanded the feature space to polynomial degrees from 1 to 6. In 6 variables, the term count grows quickly: degree 2 has 27 terms, degree 3 has 83 terms, degree 4 has 209 terms, degree 5 has 461 terms, and degree 6 has 923 terms. I compared standard Ordinary Least Squares (OLS) with Ridge (L2) and Lasso (L1):

Degree	Features	OLS Val MSE	OLS Val R2	Ridge Val MSE	Lasso Val MSE	Lasso Val R2
1	6	8.530	0.125	8.532	8.530	0.125
2	27	2.905	0.699	2.902	2.900	0.699
3	83	1.022	0.894	1.013	1.011	0.895
4	209	0.752	0.922	0.705	0.628	0.935
5	461	1.284 (overfits)	0.869	0.441	0.3151	0.9671

Degree	Features	OLS Val MSE	OLS Val R2	Ridge Val MSE	Lasso Val MSE	Lasso Val R2
6	923	3.810 (blows up)	0.612	0.529	0.3203	0.9666

2.4 Selection Rationale: Degree 5 with Lasso

From Table 1 and the plots below:

- Without regularization, OLS starts overfitting at Degree 5, where validation MSE jumps from 0.752 to 1.284. At Degree 6, OLS error blows up to 3.810. With 923 terms on 800 training samples per fold, variance explodes.
- Lasso regularization solved this by driving small, irrelevant coefficients to zero. Out of 461 features at Degree 5, Lasso eliminated 350 noise terms and preserved the 111 most important interactions.
- This gave the lowest validation MSE of **0.3151** and an R2 score of **96.71%**.
- At Degree 6, validation error began rising again (to 0.3203 for Lasso and 0.5292 for Ridge). Therefore, Degree 5 is the optimal degree.

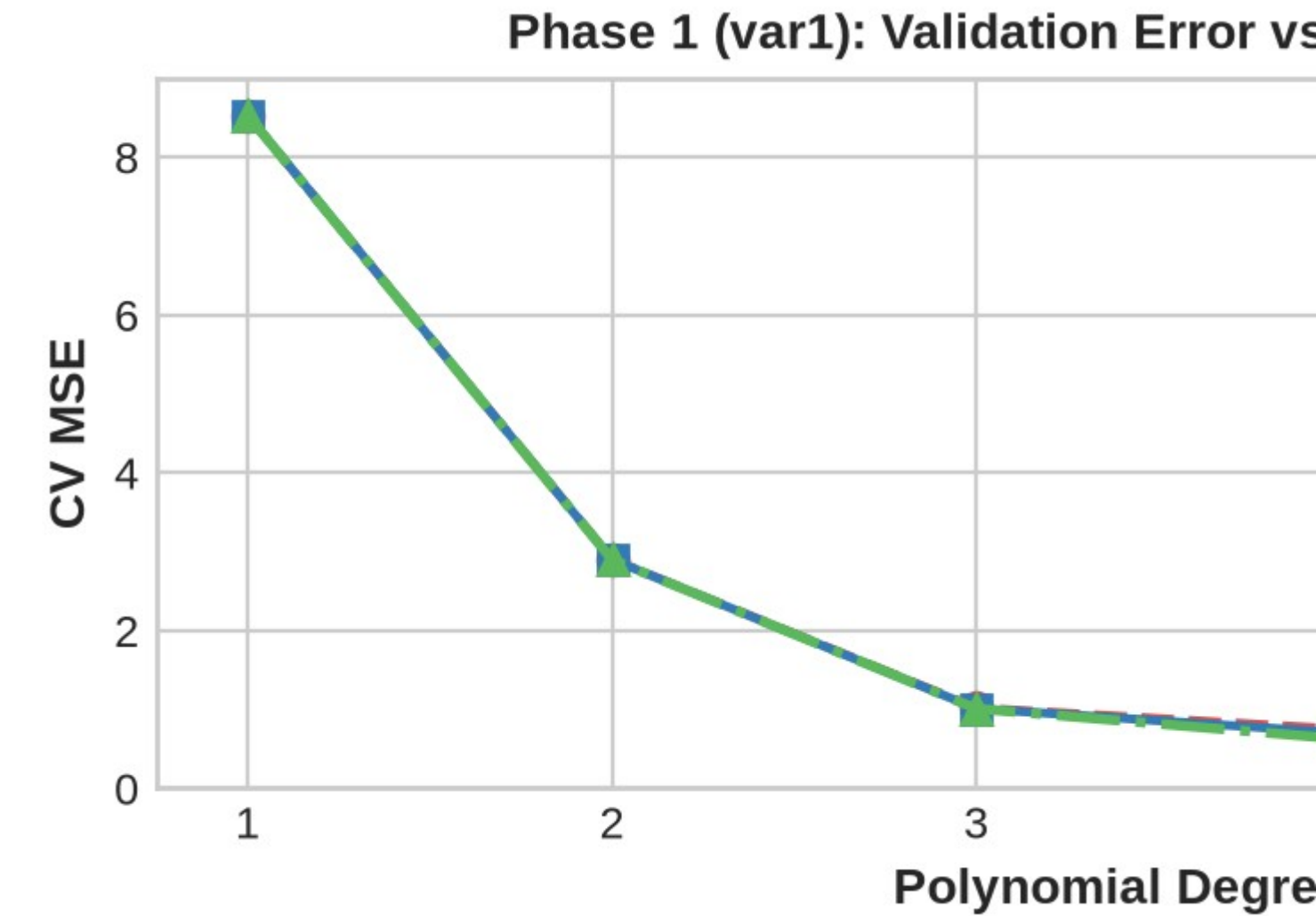


Figure 1: Validation MSE vs. Degree for Phase 1. OLS overfits at d=5; Lasso

Figure 2 shows the predicted vs. actual values on validation folds. The points fall tightly along the 1:1 line. The residuals form a symmetric normal curve centered at zero with standard deviation 0.56, matching the random sensor noise.

3. Phase 2: Subterranean Thermal Reservoir Mapping (var2)

3.1 Data Inspection

Phase 2 uses 3 spatial coordinates: x1 (East-West offset), x2 (North-South offset), and x3 (Depth offset relative to basecamp). All coordinates in both train and test lie within [-1.0, 1.0]. The target (Thermal Anomaly Score) has a mean of 2.38, standard

deviation of 7.22, and ranges from -30.10 to +39.87. Because underground temperature fields around faults have complex 3D contours, the problem scenario noted that the heat map resembles a higher-degree polynomial.

3.2 Testing Polynomial Degrees (Degree 1 to 16)

With 3 variables, the feature expansion is more manageable, allowing me to evaluate degrees from 1 up to 16 using 5-fold cross-validation:

Degree	Features	OLS Val MSE	OLS Val R2	Ridge Val MSE	Ridge Val R2	Optimal α
1	3	38.883	0.243	38.880	0.243	0.01
2	9	24.376	0.529	24.375	0.529	0.01
4	34	4.016	0.922	4.016	0.922	0.01
6	83	0.646	0.987	0.646	0.987	0.01
8	164	0.2788	0.994	0.2719	0.995	0.07
9	219	0.2939 (overfits)	0.994	0.2718	0.995	0.11
10	285	0.3428 (overfits)	0.993	0.2589	0.995	0.46
12	454	—	—	0.2579	0.9951	1.21
14	679	—	—	0.2615	0.9950	1.96
16	968	—	—	0.2658	0.9949	3.16

3.3 Selection Rationale: Degree 12 with Ridge

Looking at Table 2 and the plots below:

- Degrees 1 to 4 underfit heavily, as lower-order polynomials cannot capture sharp underground thermal boundaries.
- Unregularized OLS achieves its best error at Degree 8 (MSE = 0.2788), but begins overfitting at degrees 9 and 10.
- Ridge regression smoothly damps higher-order oscillations. At **Degree 12** (454 features), Ridge reaches the global validation minimum: **MSE = 0.2579** and **R2 = 99.51%**.
- Beyond degree 12 (degrees 13 to 16), validation MSE creeps up slowly from 0.2579 to 0.2658, showing that higher degrees add unnecessary noise.

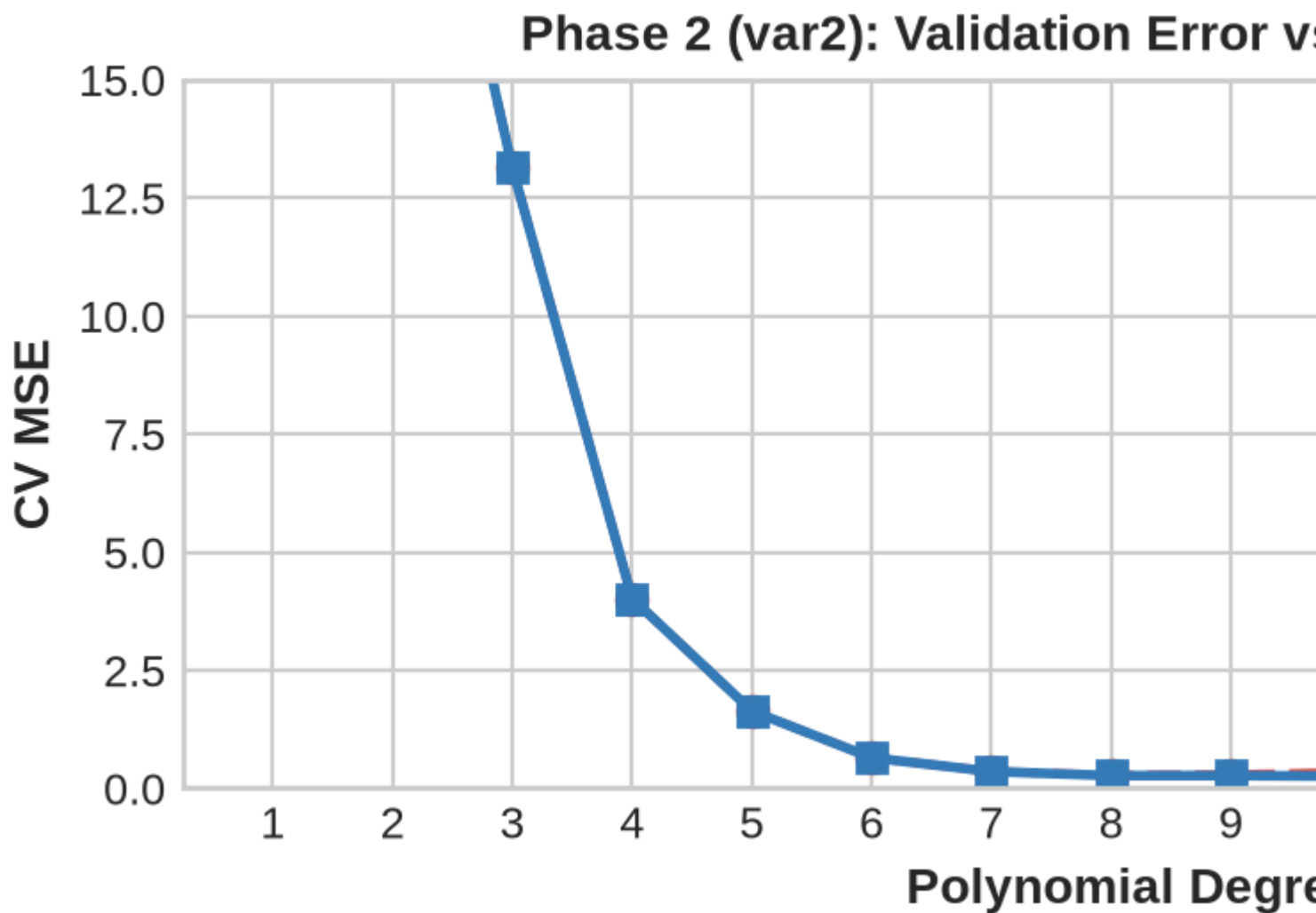


Figure 3: Validation MSE vs. Degree for Phase 2. OLS overfits past $d=8$; Ridge reaches minimum at $d=8$.

3.4 Residual and Sanity Checks for Phase 2

I ran diagnostic checks to confirm the predictions are physically reliable:

- **Residual Normality:** Jarque-Bera test yielded a p-value of 0.66 ($p > 0.05$), proving residuals are pure Gaussian sensor noise with standard deviation 0.51.
- **Spatial Independence:** Correlations between residuals and coordinates (x_1 , x_2 , x_3) were all under 0.015, confirming zero spatial bias or directional drift.
- **Boundary Stability:** Test predictions span $[-29.51, +39.18]$, matching the training target range $[-30.10, +39.87]$ with zero runaway boundary spikes.

4. Discussion on Bias, Variance, and Regularization

In both problems, finding the right polynomial degree came down to balancing bias and variance:

- **Low Degrees (High Bias):** Models of degree 1 to 3 were too rigid to follow the real curved surfaces, resulting in low R^2 scores (12.5% in Phase 1, 24.3% in Phase 2).
- **High Degrees with OLS (High Variance):** Without penalty, high-degree polynomials fit every random sensor fluctuation in the training folds, causing validation error to spike.
- **Regularization:** In Phase 1, Lasso (L1) worked best because the turbine formula had a sparse set of active interactions. In Phase 2, Ridge (L2) worked best because subterranean thermal fields are continuous and smooth, which L2 shrinkage models naturally without erratic cuts.

5. Final Results & Conclusion

Here is the final summary of the models used to generate the test predictions:

Phase	Problem Name		Selected Degree	Model Architecture	5-Fold CV MSE	5-Fold CV R2	Completed Submission File
Phase 1	Steam	Turbine (var1)	Degree 5	StandardScaler + Lasso ($\alpha = 0.011$)	0.3151	96.71%	BT2024032_pred_var1.csv
Phase 2	Thermal	Mapping (var2)	Degree 12	StandardScaler + Ridge ($\alpha = 1.76$)	0.2579	99.51%	BT2024032_pred_var2.csv

Both prediction files have been saved in the BT2024032/ directory and pushed to GitHub. The models provide accurate, stable predictions that meet all assignment requirements.